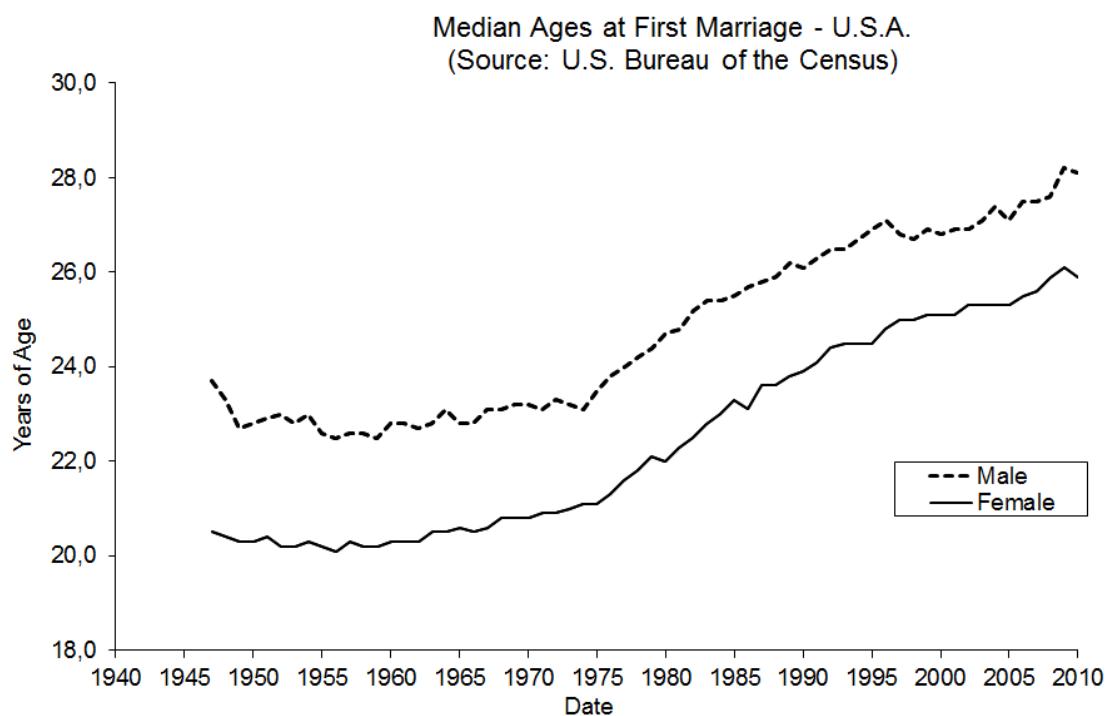


Advanced Econometrics – Early Exam - May 26, 2025**Exercise 1 (40%)**

Dave Giles at May 30, 2011, published a dataset with the estimated median age of females and males at first marriage in the USA. In the dataset median age for each year from 1947 to 2010 is stored. The blogger suggested that possible cointegration between the median age for Females (*Female*) and Males (*Male*) may exist. To assess that, several analyses were performed. Furthermore, it was decided to check whether an Error Correction Mechanism model is appropriate here. Based on the figure and results presented below please answer the following questions.



- (5%) Is the relationship between *Female* and *Male* variables real or spurious? Explain.
- (5%) Looking at the figure presented below, is there any rationality in considering the possibility that the *Female* and *Male* variables are cointegrated? Explain.
- (10%) Do time series of *Female* and *Male* variables satisfy basic assumptions to consider a cointegration between them (please support your answer with appropriate tests results)? Explain.
- (10%) Is there enough evidence to support a claim that there is cointegration between the variables (please support your answer with appropriate tests results)? Explain.
- (5%) What is or would be a cointegrating vector for the variables? How to interpret it? Explain.
- (5%) Does the ECM model work? How would you interpret coefficients in the ECM model? Explain.

Female median age at first marriage									
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4	p_bg.p.value.5		
0	1.84	0.99	0.25	0.19	0.22	0.10	0.11		
1	1.95	0.99	0.99	0.33	0.30	0.17	0.18		
2	1.29	0.99	0.77	0.94	0.80	0.70	0.44		
3	0.94	0.99	0.79	0.84	0.89	0.63	0.49		
4	0.49	0.98	0.83	0.63	0.68	0.81	0.47		
5	0.20	0.97	0.55	0.76	0.86	0.90	0.87		

First differences for female median age at first marriage									
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4	p_bg.p.value.5		
0	-8.45	0.01	0.92	0.16	0.13	0.07	0.09		
1	-4.67	0.01	0.66	0.84	0.73	0.68	0.40		
2	-3.42	0.02	0.72	0.72	0.73	0.56	0.46		
3	-2.29	0.21	0.79	0.56	0.57	0.69	0.43		
4	-2.19	0.25	0.54	0.74	0.84	0.87	0.86		
5	-1.88	0.36	0.79	0.90	0.97	0.99	0.99		

Male median age at first marriage									
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4	p_bg.p.value.5		
0	1.33	0.99	0.86	0.88	0.28	0.43	0.56		
1	1.21	0.99	0.33	0.55	0.15	0.21	0.32		
2	1.32	0.99	0.94	0.96	0.06	0.11	0.18		
3	0.82	0.99	0.92	0.94	0.89	0.95	0.98		
4	0.67	0.99	0.86	0.98	0.98	0.98	1.00		
5	0.56	0.99	0.81	0.96	0.99	0.97	0.99		

First differences for male median age at first marriage									
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4	p_bg.p.value.5		
0	-7.90	0.01	0.33	0.59	0.11	0.14	0.23		
1	-6.82	0.01	0.89	0.98	0.04	0.07	0.13		
2	-4.08	0.01	0.86	0.95	0.91	0.97	0.99		
3	-3.00	0.04	0.84	0.98	0.99	0.98	0.99		
4	-2.09	0.29	0.82	0.96	0.99	0.96	0.99		
5	-2.19	0.25	0.98	0.79	0.91	0.96	0.98		

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Results for the Long-term relationship model

```
=====
Dependent variable:
-----
Female
-----
Male                1.102***
                   (0.017)

Constant            -4.808***
                   (0.417)

-----
Observations                64
R2                          0.986
Adjusted R2                0.986
Residual Std. Error    0.249 (df = 62)
F Statistic            4,291.233*** (df = 1; 62)
=====
Note:                *p<0.1; **p<0.05; ***p<0.01
```

Residuals from Long-term relationship model								
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4	p_bg.p.value.5	
0	-5.38	0.01	0.57	0.72	0.42	0.59	0.73	
1	-4.28	0.01	0.95	0.97	0.70	0.79	0.88	
2	-2.82	0.07	0.94	0.99	0.95	0.91	0.95	
3	-2.56	0.11	0.96	1.00	1.00	0.99	0.99	
4	-2.11	0.28	0.94	0.93	0.98	0.99	0.99	
5	-2.27	0.22	0.80	0.97	0.99	0.99	0.98	

Results for the ECM model

```
=====
Dependent variable:
-----
dFemale
-----
lresid                -0.270***
                   (0.070)

dMale                0.307***
                   (0.082)

Constant            0.068***
                   (0.017)

-----
Observations                63
R2                          0.255
Adjusted R2                0.230
Residual Std. Error    0.125 (df = 60)
F Statistic            10.280*** (df = 2; 60)
=====
Note:                *p<0.1; **p<0.05; ***p<0.01
```

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KPSS test result for Male

```
#####  
# KPSS Unit Root Test #  
#####
```

Test is of type: mu with 4 lags.

Value of test-statistic is: 1.0597

Critical value for a significance level of:

```
          10pct  5pct 2.5pct  1pct  
critical values 0.347 0.463  0.574 0.739  
KPSS test results for $\Delta$ VAT$
```

KPSS test result for Δ Male

```
#####  
# KPSS Unit Root Test #  
#####
```

Test is of type: mu with 3 lags.

Value of test-statistic is: 0.0309

Critical value for a significance level of:

```
          10pct  5pct 2.5pct  1pct  
critical values 0.347 0.463  0.574 0.739
```

KPSS test result for Female

```
#####  
% # KPSS Unit Root Test #  
#####
```

Test is of type: mu with 4 lags.

Value of test-statistic is: 0.9327

% Critical value for a significance level of:

```
          10pct  5pct 2.5pct  1pct  
critical values 0.347 0.463  0.574 0.739
```

KPSS test result for Δ Female

```
#####  
# KPSS Unit Root Test #  
#####
```

Test is of type: mu with 3 lags.

Value of test-statistic is: 0.0304

Critical value for a significance level of:

```
          10pct  5pct 2.5pct  1pct  
critical values 0.347 0.463  0.574 0.739  
}
```

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Exercise 2 (30%)

In this example, we have (synthetic) data¹ on whether workers complain to managers at fast-food restaurants. The dependent variable is the dummy variable *complain* (=1 if a worker complained, =0 otherwise). The covariates are *age* (in years of the worker), *grade* (years of schooling completed by the worker), *south* (equal to 1 if the restaurant is located in the South), *tenure* (the number of years spent on the job by the worker), *gender* (of the worker, 1 for women), *race* (of the worker), *income* (in thousands of dollars by the restaurant), *genderm* (gender of the manager, 1 for women), *burger* (equal to 1 if the restaurant specializes in hamburgers), and *chicken* (equal to 1 if the restaurant specializes in chicken).²

Call:

```
glm(formula = complain ~ age + grade + south + tenure + gender +
    income, family = binomial(link = "probit"), data = chicken)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.4023853	0.5083557	0.792	0.42863
age	-0.0336174	0.0118943	-2.826	0.00471 **
grade	0.0331282	0.0243458	1.361	0.17360
south	0.0776746	0.0547301	1.419	0.15583
tenure	-0.0029834	0.0899963	-0.033	0.97355
gender	0.2975866	0.0547633	5.434	5.51e-08 ***
income	-0.0020855	0.0006771	-3.080	0.00207 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2804.7 on 2762 degrees of freedom
Residual deviance: 2752.6 on 2756 degrees of freedom
AIC: 2766.6

Call:

```
probitmfx(formula = complain ~ age + grade + south + tenure +
    gender + income, data = chicken, atmean = TRUE)
```

Marginal Effects:

	dF/dx	Std. Err.	z	P> z
age	-0.00942873	0.00333170	-2.8300	0.004655 **
grade	0.00929152	0.00682659	1.3611	0.173489
south	0.02174921	0.01529361	1.4221	0.154994
tenure	-0.00083676	0.02524139	-0.0332	0.973555
gender	0.08348932	0.01529146	5.4599	4.765e-08 ***
income	-0.00058492	0.00018957	-3.0855	0.002032 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

dF/dx is for discrete change for the following variables:

```
[1] "south" "gender"
```

¹ The data set is available at <https://www.stata-press.com/data/r17/chicken.dta>.

² The exercise is based on Stata Reference Manual (xtprobit function).

Student's book no.:

Name:

```
> null_probit = glm(complain~1, data=chicken,
family=binomial(link="probit"))
> lrtest(probit1, null_probit)
Likelihood ratio test

Model 1: complain ~ age + grade + south + tenure + gender + income
Model 2: complain ~ 1
#Df  LogLik Df Chisq Pr(>Chisq)
1    7 -1376.3
2    1 -1402.3 -6 52.11  1.773e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> gof.results = gof(probit1)
> gof.results$gof
      test  stat      val df      pVal
1:      HL chiSq 4.9805893  8 0.7596487
2:      mHL      F 0.7304362  9 0.6813310
3:      OsRo      Z 1.2660267 NA 0.2055035
4: SstPgeq0.5      Z      NaN NA      NaN
5:  SstPl0.5      Z 0.3283324 NA 0.7426603
6:  SstBoth chiSq      NaN  2      NaN
7: SllPgeq0.5 chiSq 0.0000000  1 1.0000000
8:  SllPl0.5 chiSq 0.1081492  1 0.7422611
9:  SllBoth chiSq 0.1081492  2 0.9473614
```

Use $\alpha=5\%$ significance level.

- (5\%) Interpret parameters' estimates.
- (5\%) Interpret marginal effects.
- (5\%) Are all variables jointly statistically significant? Explain.
- (7\%) Let's focus on variable *tenure*. Is it endogenous? Explain.
- (8\%) Interpret the diagnostic tests' results.

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Name:

Exercise 3 (30%)

University administrators want to know the relationship between high school grade point average (GPA) and students' performance in college. `gpa.dta` contains fictional data on a cohort of 4,000 college students.³ College GPA (`gpa2`) and high school GPA (`hsgpa`) are measured on a continuous scale between zero and four. The outcome of interest is the student's college GPA. But, for confidentiality reasons, GPAs below 2.0 are reported as 2.0. In other words, the outcome is censored on the left.

We believe that GPA is also a function of the logarithm of income of the student's parents (`pincome`) and whether or not the student participated in a study-skills program while in college (`program`).⁴

The output below presents a Tobit model estimates and the table presents marginal effects for this model.

```
Call:
censReg(formula = gpa2 ~ hsgpa + pincome + program, left = 2,
        right = Inf, data = gpa)

Observations:
      Total   Left-censored   Uncensored Right-censored
      4000             1206             2794             0

Coefficients:
              Estimate Std. error t value Pr(> |t|)
(Intercept) -0.890258    0.047848  -18.61  <2e-16 ***
hsgpa        0.658631    0.012870   51.18  <2e-16 ***
pincome      0.315930    0.007457   42.37  <2e-16 ***
program      0.555442    0.014747   37.66  <2e-16 ***
logSigma    -0.910997    0.013606  -66.95  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Newton-Raphson maximisation, 7 iterations
Return code 1: gradient close to zero
Log-likelihood: -2015.126 on 5 Df
```

```
> x = rbind(1, mean(gpa$hsgpa), mean(gpa$pincome), mean(gpa$program))
> me = tobit_marginal_effects(tobit1, x, dummies_indices=c())
```

```
Marginal effects of the tobit model
-----
              y*      E(y|x) E(y|x,y>0)      Pr(y>0|x)      at X=
hsgpa    0.6586311 0.6586311  0.6586311 8.750838e-09 2.842114
pincome  0.3159297 0.3159297  0.3159297 4.197569e-09 3.680588
program  0.5554416 0.5554416  0.5554416 7.379821e-09 0.498500
```

³ The data set is available at <https://www.stata-press.com/data/r16/gpa.dta>.

⁴ The exercise is based on Stata Reference Manual (tobit function).

Student's book no.:

Name:

Use $\alpha=5\%$ significance level.

- a) (5%) Why shouldn't we use a simple regression model for the problem? Explain.
- b) (7%) Is this a corner solution or a censoring variable case? Explain.
- c) (5%) What is the interval of values of the dependent variable? How many observations did take values on the edge of the interval? Explain.
- d) (5%) Which variables are individually significant? Explain.
- e) (8%) What is the effect of participation in a study-skills program while in college? Use appropriate marginal effect. Explain your answer.

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